

5 MW Wind Turbine Multi-Domain Control System

This repository contains a dual-domain MATLAB/Simulink model for a 5 MW wind turbine. The project evaluates advanced non-linear control strategies for both the mechanical (aerodynamic) and electrical (generator) subsystems to ensure stable operation and power regulation.

System Architecture

The model is organized into five functional stages, reflecting a complete control loop from wind input to grid-side electrical variables:

1. Wind Input: Offers constant, step, or turbulent wind profiles.
2. Aerodynamic Controller Choice: Implements pitch control strategies (β) for Region 3 operation.
3. Aerodynamic Plant: Models the rotor dynamics ($\dot{\omega}$) using an affine $F + G \cdot u$ structure.
4. Electrical Plant Controller Choice: Manages the Permanent Magnet Synchronous Generator (PMSG) currents (i_d , i_q).
5. Electrical Plant: Simulates the scalar PMSG dynamics and electrical torque generation.

Control Strategies

The system allows for independent testing of four non-linear control algorithms across two domains:

Aerodynamic Domain (Pitch Control)

Controller	Primary Advantage
Input-Output (I/O) Linearization	Simplifies non-linear turbine physics into a linear tracking problem.
Sliding Mode Control (SMC)	Provides high robustness against severe wind gusts and model uncertainties.
Dynamic SMC	Smooths the control signal to reduce mechanical stress on the pitch actuators.
Super-Twisting SMC	Offers second-order precision with minimal chattering for high-fidelity regulation.

Electrical Domain (Generator Control)

- I/O Linearization: Decouples the d and q axis currents for precise torque control.

- SMC Controller: Ensures fast current response and rejection of electrical disturbances.
- Adaptive Super-Twisting (ASTW) SMC: The most advanced option, adapting gains in real-time to optimize efficiency and minimize losses without needing exact system parameters.

Mathematical Implementation: Plant_Q1_Simple

The aerodynamic core of the simulation is a MATLAB function that represents the rotor acceleration in affine form:

$$\dot{\omega} = F(\omega, v) + G(\omega, v) \cdot u + \delta$$

- Affine Structure: By breaking the physics into F (internal dynamics) and G (control gain), we can apply non-linear cancellation techniques.
- Disturbance (δ): The model includes a disturbance port to test the robustness of the SMC and ASTW controllers against noise and modeling errors.

Key Features & Tools

- w_star Estimator: A specialized function block used to calculate the Rotor-Effective Wind Speed, enabling feedforward control to anticipate wind changes.
- PMSG Scalar Plant: Models the electrical side including Vd and Vq voltages to regulate the generator's electromagnetic torque.
- Multi-Scope Monitoring: Real-time tracking of Generator Speed (Ω), Power Output, and d-q current tracking (i_q vs i_q^*).

How to Use

1. Open and run the wind_s.m file.
2. Open the .slx file
3. Select Wind: Choose the wind type in the green block.
4. Enable Aerodynamic Controller: Click the variant or switch in the "Aerodynamic Controller Choice" block.
5. Enable Electrical Controller: Select the corresponding current controller in the "Electrical Plant Controller Choice" block.

6. Analyze: Run the simulation and use the scopes to compare tracking performance and chattering levels between the different algorithms.